

|     |                                                  |
|-----|--------------------------------------------------|
| 001 | <b>0.1 Prompt Templates</b>                      |
| 002 | The prompt template for both players of the Deal |
| 003 | or No Deal is given in Figure 1.                 |
| 004 | <b>0.2 Experimental Setup</b>                    |
| 005 | <b>0.3 Overall Results</b>                       |
| 006 | <b>0.4 Detailed Analysis</b>                     |
| 007 | <b>0.5 Qualitative Samples</b>                   |

#### TEMPLATE 0.1.1

You are given three images, one is called target and the other two are distractors. Your task is to generate a referring expression that best describes the target image while distinguishing it from the two other distractor images. The first image is <IMAGE\_POSITION>, the second image is <IMAGE\_POSITION>, and the third image is <IMAGE\_POSITION>.

Instruction: Describe the target image. Generate the referring expression starting with the tag "Expression: " for the given target image. Omit any other text.

Target image: <IMAGE\_PATH>  
Second image: <IMAGE\_PATH>  
Third image: <IMAGE\_PATH>

#### TEMPLATE 0.1.3

You are given three images. You are also given a referring expression that describes one of the given images. Your task is to select the image that matches the given referring expression. Generate only the number (in text) of the image that matches the given expression by selecting first, second, or third.

#### TARGET\_EXPRESSION

Question: Which image does the expression refer to? Start with the tag "Answer: ", followed by your selection. Omit any other text.

First image: <IMAGE\_PATH>  
Second image: <IMAGE\_PATH>  
Third image: <IMAGE\_PATH>

#### TEMPLATE 0.1.2

Expression: \$EXPRESSION\$

#### TEMPLATE 0.1.4

Answer: \$ANSWER\$

(a) Prompt template for Player A (Instruction Giver) in the Reference Game.

(b) Prompt template for Player B (Instruction Follower) in the Reference Game

Figure 1: Reference game prompt templates for both players

|            |                         | Claude<br>Reason | No reason | Deepseek<br>r1-distilled | v3 | GPT5<br>Reason | No reason | GPT5-mini<br>Reason | No reason | GPT-OSS | LLAMA | Nemotron<br>Reason | No reason | Qwen3<br>Thinking |
|------------|-------------------------|------------------|-----------|--------------------------|----|----------------|-----------|---------------------|-----------|---------|-------|--------------------|-----------|-------------------|
| Optimal    | Explicit maximization   | 0                | 1         | 0                        | 3  | 11             | 3         | 11                  | 0         | 1       | 0     | 0                  | 0         | 5                 |
|            | Simple negotiation      | 17               | 16        | 0                        | 7  | 3              | 2         | 6                   | 2         | 5       | 5     | 0                  | 0         | 8                 |
|            | Luck                    | 0                | 0         | 0                        | 0  | 0              | 0         | 0                   | 0         | 0       | 0     | 1                  | 0         | 0                 |
|            | Unclear preferences     | 1                | 1         | 1                        | 1  | 0              | 1         | 0                   | 0         | 1       | 1     | 0                  | 0         | 0                 |
| Suboptimal | Unclaimed items         | 0                | 0         | 0                        | 0  | 0              | 0         | 0                   | 0         | 0       | 1     | 0                  | 0         | 1                 |
|            | Claiming valueless item | 1                | 2         | 0                        | 0  | 0              | 2         | 0                   | 1         | 3       | 4     | 0                  | 0         | 0                 |
|            | Wrong proposal          | 0                | 0         | 1                        | 0  | 0              | 0         | 0                   | 0         | 1       | 0     | 0                  | 0         | 0                 |
|            | Premature proposal      | 0                | 0         | 3                        | 1  | 3              | 0         | 1                   | 0         | 3       | 0     | 3                  | 1         | 0                 |
| Failed     | Wrong proposal          | 0                | 0         | 3                        | 7  | 1              | 11        | 0                   | 13        | 1       | 5     | 0                  | 0         | 4                 |
|            | Premature proposal      | 1                | 0         | 12                       | 1  | 1              | 0         | 1                   | 2         | 4       | 0     | 14                 | 19        | 0                 |
| Aborted    | Wrong syntax            | 0                | 0         | 0                        | 0  | 0              | 1         | 0                   | 1         | 1       | 4     | 0                  | 0         | 1                 |
|            | Non existant items      | 0                | 0         | 0                        | 0  | 0              | 0         | 0                   | 0         | 0       | 0     | 2                  | 0         | 0                 |
|            | No proposal             | 0                | 0         | 0                        | 0  | 0              | 0         | 0                   | 0         | 0       | 0     | 0                  | 0         | 1                 |

Table 1: Game outcomes for the cooperative DoND game mode.

|            |                         | Claude<br>Reason | No reason | Deepseek<br>r1-distilled | v3 | GPT5<br>Reason | No reason | GPT5-mini<br>Reason | No reason | GPT-OSS | LLAMA | Nemotron<br>Reason | No reason | Qwen3<br>Thinking |
|------------|-------------------------|------------------|-----------|--------------------------|----|----------------|-----------|---------------------|-----------|---------|-------|--------------------|-----------|-------------------|
| Optimal    | Explicit maximization   | 0                | 1         | 0                        | 1  | 0              | 0         | 0                   | 0         | 0       | 0     | 0                  | 0         | 0                 |
|            | Simple negotiation      | 18               | 13        | 1                        | 6  | 14             | 3         | 9                   | 1         | 6       | 5     | 0                  | 0         | 6                 |
|            | Luck                    | 0                | 0         | 2                        | 0  | 0              | 0         | 0                   | 0         | 2       | 0     | 2                  | 0         | 0                 |
|            | Unclear preferences     | 0                | 0         | 0                        | 1  | 0              | 0         | 0                   | 0         | 0       | 0     | 0                  | 0         | 0                 |
| Suboptimal | Unclaimed items         | 0                | 0         | 0                        | 0  | 1              | 1         | 2                   | 5         | 0       | 1     | 0                  | 0         | 3                 |
|            | Claiming valueless item | 1                | 3         | 0                        | 5  | 2              | 1         | 2                   | 3         | 0       | 2     | 0                  | 0         | 2                 |
|            | Wrong proposal          | 0                | 0         | 0                        | 0  | 0              | 0         | 0                   | 0         | 0       | 0     | 0                  | 0         | 0                 |
|            | Premature proposal      | 0                | 0         | 1                        | 0  | 2              | 0         | 0                   | 0         | 0       | 0     | 2                  | 1         | 0                 |
| Failed     | Wrong proposal          | 0                | 3         | 5                        | 6  | 1              | 12        | 6                   | 8         | 7       | 2     | 0                  | 0         | 2                 |
|            | Premature proposal      | 1                | 0         | 7                        | 0  | 0              | 1         | 1                   | 1         | 4       | 1     | 14                 | 18        | 6                 |
| Aborted    | Wrong syntax            | 0                | 0         | 4                        | 1  | 0              | 0         | 0                   | 2         | 1       | 9     | 2                  | 0         | 1                 |
|            | Non existant items      | 0                | 0         | 0                        | 0  | 0              | 0         | 0                   | 0         | 0       | 0     | 1                  | 1         | 0                 |
|            | No proposal             | 0                | 0         | 0                        | 0  | 0              | 0         | 0                   | 0         | 0       | 0     | 0                  | 0         | 0                 |

Table 2: Game outcomes for the semi-competitive DoND game mode.

|                   |                           | Claude<br>Reason | No reason | Deepseek<br>r1-distilled | v3 | GPT5<br>Reason | No reason | GPT5-mini<br>Reason | No reason | GPT-OSS | LLAMA | Nemotron<br>Reason | No reason | Qwen3<br>Thinking | Instruct |
|-------------------|---------------------------|------------------|-----------|--------------------------|----|----------------|-----------|---------------------|-----------|---------|-------|--------------------|-----------|-------------------|----------|
| Tell exact values | Tell exact values         | 0                | 3         | 1                        | 13 | 19             | 19        | 17                  | 12        | 14      | 1     | 0                  | 0         | 13                | 17       |
|                   | Tell only relative values | 20               | 17        | 0                        | 5  | 0              | 1         | 0                   | 7         | 0       | 18    | 0                  | 0         | 2                 | 1        |
|                   | Do not tell preferences   | 0                | 0         | 7                        | 2  | 0              | 0         | 2                   | 0         | 5       | 1     | 0                  | 0         | 5                 | 0        |
|                   | Do not communicate at all | 0                | 0         | 12                       | 0  | 0              | 0         | 0                   | 0         | 1       | 0     | 20                 | 20        | 0                 | 2        |

Table 3: Model willingness to share its value function for the cooperative DoND game mode.

|                           | Claude |           | Deepseek     |    | GPT5   |           | GPT5-mini |           | GPT-OSS | LLAMA | Nemotron |           | Qwen3    | Instruct |
|---------------------------|--------|-----------|--------------|----|--------|-----------|-----------|-----------|---------|-------|----------|-----------|----------|----------|
|                           | Reason | No reason | r1-distilled | v3 | Reason | No reason | Reason    | No reason |         |       | Reason   | No reason | Thinking |          |
| Tell exact values         | 0      | 1         | 0            | 11 | 0      | 4         | 0         | 9         | 5       | 0     | 0        | 0         | 6        | 6        |
| Tell only relative values | 19     | 19        | 1            | 9  | 15     | 14        | 8         | 11        | 7       | 20    | 0        | 0         | 1        | 10       |
| Do not tell preferences   | 1      | 0         | 8            | 0  | 5      | 0         | 11        | 0         | 7       | 0     | 0        | 0         | 12       | 0        |
| Do not communicate at all | 0      | 0         | 11           | 0  | 0      | 0         | 1         | 0         | 1       | 0     | 20       | 20        | 1        | 4        |

Table 4: Model willingness to share its value function for the semi-competitive DoND game mode.

|                            | Claude |           | Deepseek     |    | GPT5   |           | GPT5-mini |           | GPT-OSS | LLAMA | Nemotron |           | Qwen3    | Instruct |
|----------------------------|--------|-----------|--------------|----|--------|-----------|-----------|-----------|---------|-------|----------|-----------|----------|----------|
|                            | Reason | No reason | r1-distilled | v3 | Reason | No reason | Reason    | No reason |         |       | Reason   | No reason | Thinking |          |
| Lie about preferences      | 1      | 1         | 0            | 9  | 0      | 0         | 0         | 1         | 0       | 2     | 0        | 0         | 0        | 0        |
| Dont lie about preferences | 19     | 19        | 20           | 11 | 19     | 20        | 19        | 18        | 20      | 18    | 20       | 20        | 20       | 20       |

Table 5: Model truthfulness about its value function for the cooperative DoND game mode.

|                            | Claude |           | Deepseek     |    | GPT5   |           | GPT5-mini |           | GPT-OSS | LLAMA | Nemotron |           | Qwen3    | Instruct |
|----------------------------|--------|-----------|--------------|----|--------|-----------|-----------|-----------|---------|-------|----------|-----------|----------|----------|
|                            | Reason | No reason | r1-distilled | v3 | Reason | No reason | Reason    | No reason |         |       | Reason   | No reason | Thinking |          |
| Lie about preferences      | 0      | 5         | 0            | 6  | 0      | 0         | 0         | 2         | 1       | 3     | 0        | 0         | 0        | 1        |
| Dont lie about preferences | 20     | 15        | 20           | 14 | 20     | 18        | 20        | 18        | 19      | 17    | 20       | 20        | 20       | 0        |

Table 6: Model truthfulness about its value function for the semi-competitive DoND game mode.